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SPECIFIC COMMENTS

Manuscript Title: A Comprehensive Review of Environmental and Nutritional Influences on Epigenetic Change
Manuscript Type: Narrative Review
Overall Recommendation: Minor Revision

Reviewer:

General Assessment

This manuscript presents a comprehensive and timely narrative review of nutritional, environmental, and lifestyle influences on epigenetic regulation across the lifespan. The authors integrate molecular mechanisms (DNA methylation, histone modifications, non-coding RNAs, and epitranscriptomics) with translational implications for chronic disease, preventive medicine, and personalized nutrition. The scope is broad yet generally well organized, and the reference list is current and extensive.

The manuscript is strongest in its mechanistic integration and translational framing. Minor revisions are recommended to strengthen academic rigor, clarify evidentiary weight, and improve balance for a broad biomedical readership.

Minor Comments

Abstract: Consider adding a brief concluding sentence summarizing the central take-home message and translational relevance.

Table 1: The table is clear but could be enhanced by adding a column indicating primary epigenetic mechanisms involved (DNA methylation, histone modification, ncRNAs).

Section Transitions: Short integrative transition paragraphs between major sections (e.g., mechanisms → nutrition → environment) would improve narrative flow.

Terminology: Ensure consistent use of terms such as “epigenetic reprogramming,” “epigenetic drift,” and “metabolic memory.”

Final Recommendation

Minor Revision

The manuscript is scientifically sound, well written, and makes a valuable contribution to the field of nutritional and environmental epigenetics. Addressing the points above will further strengthen clarity, balance, and suitability for publication in a high-impact review journal.

Plagiarism report Enclosed.